

GA1 Project: Building and Validating an SDN-Driven Virtual Network in GNS3.

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Date:	23/10/2025

Leave Empty for examiner:

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EXAMINER NAME	Dr Angus Brandt
INTERNAL MODERATOR	Dr Ali Almaktoof
EXTERNAL MODERATOR	Mr Shamil Talip

Submit:

Thursday, 16 October 2025, Midnight on Blackboard.

PDF only. No zipped files or Word documents.

Demonstration:

Friday 17 October, 9 AM (demo during class time is critical (9 AM – 12 PM)).

All students must report to class at 9 AM.

The demonstration validates the report.

No late demonstrations will be accepted.

Metadata (Complete with your details)

Student Number:

LAST_DIGIT:

SECOND LAST DIGIT:

THIRD LAST DIGIT:

ROUTER COUNT: (Allowed: 4–7)

PROTOCOL: (OSPF/EIGRP)

UNIQUE Topology:

1. Introduction

Enterprise networks increasingly require migration from traditional distributed routing to centralized Software-Defined Networking (SDN) architectures while maintaining service continuity and network stability. This project addresses the migration challenge by integrating a modern IS-IS backbone network with an OpenDaylight SDN controller using BGP as the control-plane bridge. The problem is broadly defined because it encompasses multiple architectural layers: network topology design with resilient multi-path connectivity, protocol interoperability between legacy link-state routing (IS-IS) and modern SDN control planes, IPv4 addressing schemes across redundant links, controller integration with proper BGP session establishment, and comprehensive validation mechanisms to verify network correctness.

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2. Problem Definition (GA1 – Indicator 1)

Define the migration challenge clearly, identify constraints (router limit, addressing, protocol, SDN integration).

Migration Challenge: An enterprise operates a traditional IS-IS backbone network with four routers arranged in a square topology with diagonal cross-connections, providing resilience and multi-path routing. The network requires SDN controller integration for centralized policy management, network telemetry, and dynamic route control without disrupting existing IS-IS adjacencies.

Constraints Identified:

- **Router Limit:** 4 VyOS routers (VyOS1, VyOS2, VyOS3, VyOS4) within GNS3 constraints

- **Topology:** Square perimeter (4 primary links) + 2 diagonal cross-connections = 6 total links for redundancy
- **Addressing Scheme:** /30 subnets for point-to-point links (10.1.x.0/30 range); /32 loopbacks for stable IS-IS router-IDs (1.1.1.1 through 4.4.4.4)
- **IS-IS Configuration:** Level-2 routing, all routers in single area (49.0001), Network Entity Titles (NETs) configured
- **BGP/SDN Integration:** iBGP session from VyOS1 (192.168.56.100) to OpenDaylight controller (192.168.56.105), AS 65002
- **Port Binding:** ODL listening on port 1790 for BGP peer acceptance

3. Formulate Strategies (GA1 – Indicator 2)

Compare static vs dynamic routing, justify your chosen protocol, and present alternatives with pros/cons.

Aspect	Static Routing	IS-IS (Dynamic)
Scalability	Poor - manual config per route	Excellent - automatic discovery
Failover	None - manual intervention	Automatic via SPF convergence
Multi-path Support	Limited	Full ECMP support
Redundancy	Manual path specification	Automatic via multiple links
Management Overhead	High for complex topologies	Low - self-learning
Adaptability	Rigid	Responsive to topology changes

Chosen Protocol Justification:

IS-IS for Backbone: Selected because it provides automatic link discovery, fast convergence (<1 second), loop prevention via SPF algorithm, and no routing loops. The square topology with diagonals creates natural redundancy that IS-IS fully exploits. IS-IS is vendor-neutral and widely supported across enterprise equipment.

iBGP for SDN Integration: Allows direct communication between VyOS backbone and OpenDaylight controller. BGP's policy framework enables future route filtering, prepending, and advanced SDN-driven path selection. iBGP maintains separation between internal topology (IS-IS) and external control plane (SDN).

Alternatives and Pros/Cons:

1. OSPF Instead of IS-IS

- Pros: More commonly known, easier initial configuration, integrated route summarization
- Cons: Slower convergence, less suitable for large-scale deployments, flooding overhead
- Decision: Rejected; IS-IS chosen for better performance and SDN alignment

2. Direct ODL Integration Without BGP

- Pros: Simplified protocol stack
- Cons: ODL lacks native IS-IS support; requires translation layer; loses BGP policy benefits
- Decision: Not viable; BGP required for industry-standard SDN control

3. eBGP Instead of iBGP

- Pros: Can simulate AS boundaries, more realistic WAN simulation
- Cons: Unnecessary complexity for lab; requires AS renumbering; complicates IS-IS interaction
- Decision: iBGP chosen for simplicity and direct controller coupling

4. Analysis & Evaluation (GA1 – Indicator 3)

Provide IP addressing plan, routing adjacency tables, and BGP session evidence.

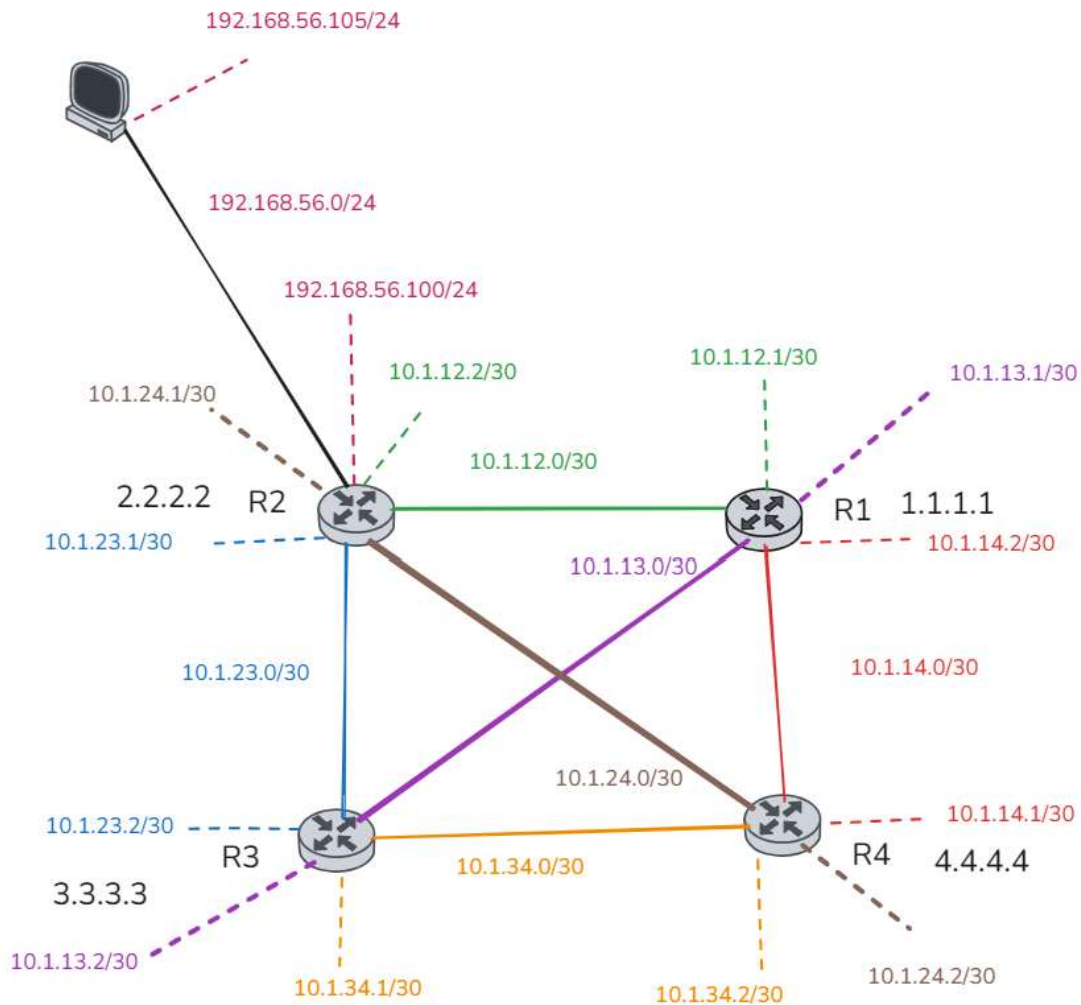


Figure 1: IP addressing plan visualization

IP Addressing Plan:

Device	IP Address	Link	Interface	Purpose
VyOS1	1.1.1.1/32	Lo	Loopback	IS-IS Router-ID
VyOS1	10.1.12.1/30	Link1	eth0	VyOS1-VyOS2
VyOS1	10.1.14.2/30	Link4	eth1	VyOS4-VyOS1
VyOS1	10.1.13.1/30	Link5	eth2	VyOS1-VyOS3 (Diagonal)
VyOS2	2.2.2.2/32	Lo	Loopback	IS-IS Router-ID
VyOS2	10.1.12.2/30	Link1	eth0	VyOS1-VyOS2
VyOS2	10.1.23.1/30	Link2	eth1	VyOS2-VyOS3
VyOS2	10.1.24.1/30	Link6	eth2	VyOS2-VyOS4 (Diagonal)
VyOS3	3.3.3.3/32	Lo	Loopback	IS-IS Router-ID
VyOS3	10.1.23.2/30	Link2	eth0	VyOS2-VyOS3
VyOS3	10.1.34.1/30	Link3	eth1	VyOS3-VyOS4
VyOS3	10.1.13.2/30	Link5	eth2	VyOS1-VyOS3 (Diagonal)
VyOS4	4.4.4.4/32	Lo	Loopback	IS-IS Router-ID
VyOS4	10.1.34.2/30	Link3	eth0	VyOS3-VyOS4
VyOS4	10.1.14.1/30	Link4	eth1	VyOS4-VyOS1
VyOS4	10.1.24.2/30	Link6	eth2	VyOS2-VyOS4 (Diagonal)
ODL	192.168.56.105/24	Mgmt	eth0	SDN Controller
VyOS1	192.168.56.100/24	Mgmt	Ext.	BGP to ODL

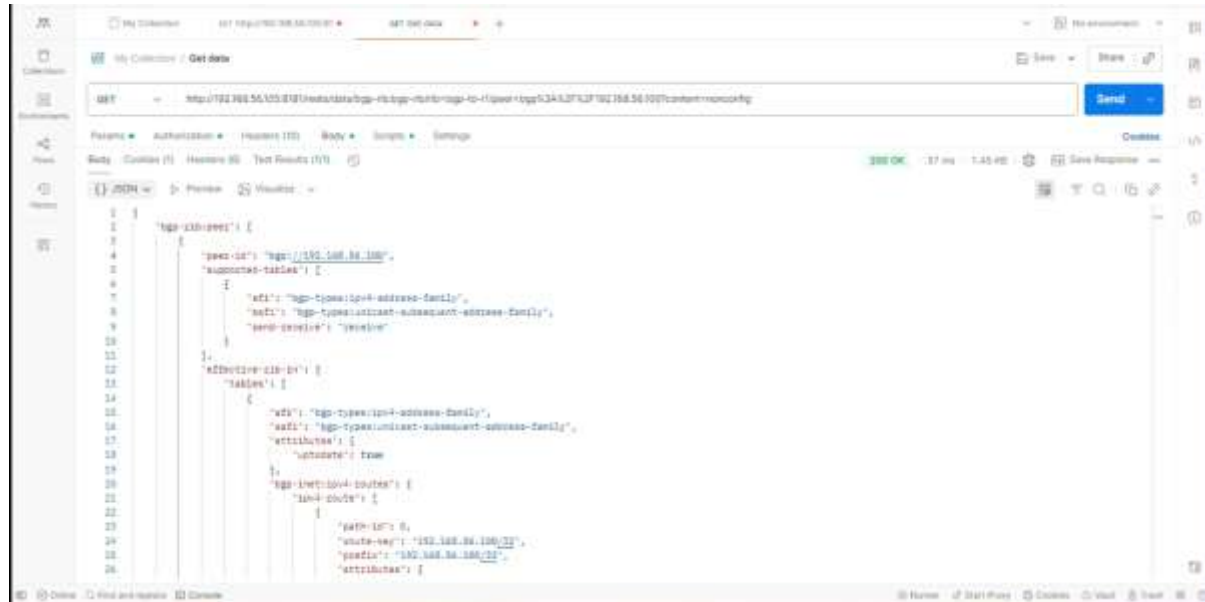


Figure 2 POSTMAN_BGP – Insert screenshot of Postman showing BGP session 1.

5. Implementation (GA1 – Indicator 4)

Provide VyOS configs, switch config, Postman BGP setup, and Python SDN application details.

VyOS Interface Configurations

VyOS1 Interfaces:

configure

```
set interfaces ethernet eth0 address 10.1.12.1/30
```

```
set interfaces ethernet eth1 address 10.1.14.2/30
```

```
set interfaces ethernet eth2 address 10.1.13.1/30
```

```
set interfaces loopback lo address 1.1.1.1/32
```

commit

save

exit

```
vyos@vyos:~$ show interfaces
Codes: S - State, L - Link, u - Up, D - Down, A - Admin Down
Interface    IP Address    MAC                VRF    MTU    S/L    Description
-----
eth0         10.1.12.1/30  08:00:27:42:0e:0b  default 1500   u/u
eth1         10.1.14.2/30  08:00:27:d4:18:50  default 1500   u/u
eth2         10.1.13.1/30  08:00:27:b5:df:b1  default 1500   u/u
eth3         -             08:00:27:49:4b:21  default 1500   u/D
lo           127.0.0.1/8   00:00:00:00:00:00  default 65536  u/u
             1.1.1.1/32
             ::1/128
vyos@vyos:~$
```

VyOS2 Interfaces:

configure

set interfaces ethernet eth0 address 10.1.12.2/30

set interfaces ethernet eth1 address 10.1.23.1/30

set interfaces ethernet eth2 address 10.1.24.1/30

set interfaces loopback lo address 2.2.2.2/32

commit

save

exit

```
vyos@vyos:~$ show interfaces
Codes: S - State, L - Link, u - Up, D - Down, A - Admin Down
Interface    IP Address    MAC                VRF    MTU    S/L    Description
-----
eth0         10.1.12.2/30  08:00:27:58:c6:7a  default 1500   u/u
eth1         10.1.23.1/30  08:00:27:d4:18:50  default 1500   u/u
eth2         10.1.24.1/30  08:00:27:6d:a2:c4  default 1500   u/u
eth3         192.168.56.100/24  08:00:27:0d:91:a3  default 1500   u/u
lo           127.0.0.1/8     00:00:00:00:00:00  default 65536  u/u
             2.2.2.2/32
             192.168.56.100/32
             ::1/128
vyos@vyos:~$
```

VyOS3 Interfaces:

configure

set interfaces ethernet eth0 address 10.1.23.2/30

set interfaces ethernet eth1 address 10.1.34.1/30

set interfaces ethernet eth2 address 10.1.13.2/30

set interfaces loopback lo address 3.3.3.3/32

commit

save

exit

```
exit
vyos@vyos:~$ show interfaces
Codes: S - State, L - Link, u - Up, D - Down, A - Admin Down
Interface  IP Address  MAC  VRF  MTU  S/L  Description
-----
eth0       10.1.23.2/30  08:00:27:54:e9:a5  default  1500  u/u
eth1       10.1.34.1/30  08:00:27:d4:18:50  default  1500  u/u
eth2       10.1.13.2/30  08:00:27:c2:c2:f7  default  1500  u/u
eth3       -            08:00:27:7d:b7:a7  default  1500  u/D
lo         127.0.0.1/8   00:00:00:00:00:00  default  65536 u/u
          3.3.3.3/32
          ::1/128
vyos@vyos:~$
```

VyOS4 Interfaces:

configure

set interfaces ethernet eth0 address 10.1.34.2/30

set interfaces ethernet eth1 address 10.1.14.1/30

set interfaces ethernet eth2 address 10.1.24.2/30

set interfaces loopback lo address 4.4.4.4/32

commit

save

exit

```
exit
vyos@vyos:~$ show interfaces
Codes: S - State, L - Link, u - Up, D - Down, A - Admin Down
Interface    IP Address    MAC                VRF    MTU    S/L    Description
-----
eth0         10.1.23.2/30   08:00:27:54:e9:a5  default 1500   u/u
eth1         10.1.34.1/30   08:00:27:d4:18:50  default 1500   u/u
eth2         10.1.13.2/30   08:00:27:c2:c2:f7  default 1500   u/u
eth3         -              08:00:27:7d:b7:a7  default 1500   u/D
lo           127.0.0.1/8    00:00:00:00:00:00  default 65536  u/u
             3.3.3.3/32
             ::1/128
vyos@vyos:~$
```

IS-IS Protocol Configuration

All VyOS Routers (VyOS1-4) IS-IS Setup:

configure

set protocols isis net 49.0001.0000000000001.00 # VyOS1 (change last 0001 to 0002/0003/0004 for others)

set protocols isis level level-2

set protocols isis interface eth0

set protocols isis interface eth1

set protocols isis interface eth2

set protocols isis interface lo

commit

save

exit

BGP Configuration (VyOS1 to ODL)

VyOS1 iBGP to OpenDaylight:

configure

```
set protocols bgp system-as 65002
```

```
set protocols bgp parameters router-id 192.168.56.100
```

```
set protocols bgp neighbor 192.168.56.105 description 'ODL iBGP'
```

```
set protocols bgp neighbor 192.168.56.105 remote-as 65002
```

```
set protocols bgp neighbor 192.168.56.105 timers connect 10
```

```
set protocols bgp neighbor 192.168.56.105 timers holdtime 90
```

```
set interfaces loopback lo address 192.168.56.100/32
```

```
set protocols bgp address-family ipv4-unicast
```

```
set protocols bgp neighbor 192.168.56.105 address-family ipv4-unicast
```

```
set protocols bgp address-family ipv4-unicast network 192.168.56.100/32
```

```
set protocols bgp neighbor 192.168.56.105 port 1790
```

commit

save

exit

Postman BGP Setup

8-Step Postman Sequence (see FIG: POSTMAN_BGP for screenshot):

1. **GET** - Check default policies
2. **GET** - Check RIB config
3. **POST** - Create BGP protocol instance (bgp-to-r1)

4. **GET** - Verify protocol instance
5. **PUT** - Configure port binding (1790)
6. **POST** - Add VyOS1 as neighbor
7. **GET** - Verify neighbor connection (routes visible)
8. **PUT** - Configure route reflector

All URLs use 192.168.56.105:8181 with Basic Auth (admin/admin)

Python SDN Application

Python script queries ODL REST API to extract:

- Network topology and instances
- BGP peer information
- Received routes with origin/next-hop
- Local RIB summary

Usage: python odl_sdn_app.py

```
=====
OPENDAYLIGHT SDN NETWORK ANALYSIS APPLICATION
Student ID: 222170972
=====

[1/5] Connecting to OpenDaylight Controller...
Raw network data saved to: network_data.json

[2/5] Analysing BGP Information...
Found 1 BGP peer(s) and 0 route(s)

[3/5] Building Network Topology...
Topology built with 0 routers and 0 links

[4/5] Generating Network Analysis Report...

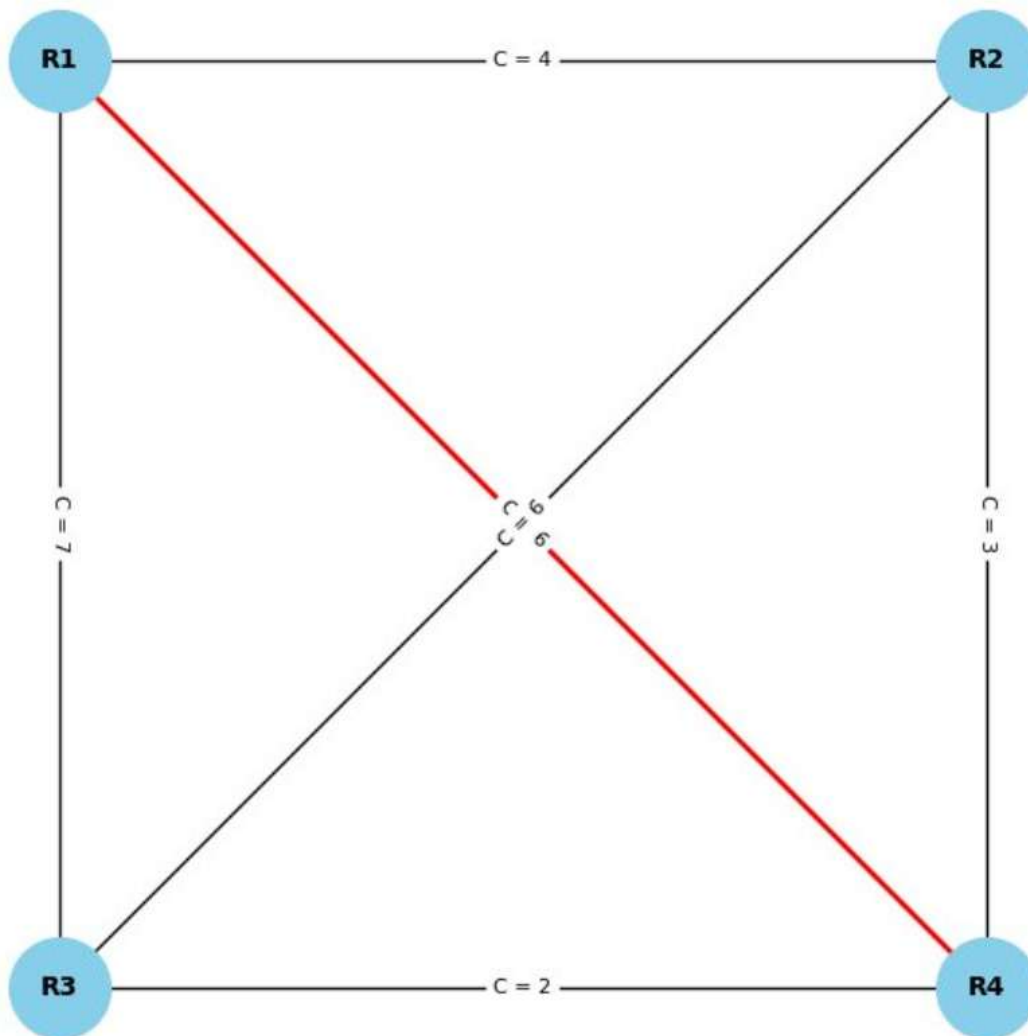
=====
BGP NEIGHBOUR INFORMATION
=====

Total BGP Neighbours: 1

--- Neighbour 1 ---
Peer ID:      bgp://192.168.56.100
Peer Role:    rr-client

Supported Address Families:
- bgp-types:ipv4-address-family / bgp-types:unicast-subsequent-address-family

=====
BGP ROUTING INFORMATION
=====
No routes found
```



6. Validation & Results

Show ping/traceroute results and confirm Python SDN application outputs match the actual network.

IS-IS Verification using router 2sh

```
vyos@vyos:~$ show isis database
Area Vy05:
IS-IS Level-2 link-state database:
LSP ID          PduLen  SeqNumber  Chksum  Holdtime  ATT/P/OI
vyos.00-00      *    124    0x0000000a  0x91d1    563    0/0/0
vyos.00-00      144    0x0000000a  0x7779    574    0/0/0
vyos.02-00       51    0x00000006  0x6d56    874    0/0/0
vyos.03-00       51    0x00000006  0x8e32    850    0/0/0
vyos.00-00      124    0x001218cc  0x38fb    547    0/0/0
vyos.04-00       51    0x00110df4  0x179b    798    0/0/0
6 LSPs
vyos@vyos:~$
```

```
vyos@vyos:~$ show isis summary
vrf          : default
Process Id   : 1331
System Id    : 0000.0000.0001
Up time      : 01:20:21 ago
Number of areas : 1
Area Vy05:
  Net: 49.0001.0000.0000.0001.00
  TX counters per PDU type:
    L2 IIH: 5079
    L2 LSP: 6630524
    L2 PSNP: 1
    LSP RXMT: 0
  RX counters per PDU type:
    L2 IIH: 5042
    L2 LSP: 7383315
    L2 CSNP: 1068
  Drop counters per PDU type:
    L2 LSP: 539881
  Advertise high metrics: Disabled
Level-2:
  LSP0 regenerated: 10
  LSPs purged: 144285
SPF: (pending)
  minimum interval : 1
IPv4 route computation:
  last run elapsed : 00:00:01 ago
  last run duration : 126 usec
  run count        : 3390
IPv6 route computation:
  last run elapsed : 00:00:01 ago
  last run duration : 27 usec
  run count        : 3390
```

BGP Verification

```
vyos@vyos:~$ nc -vz 192.168.56.105 1790
Connection to 192.168.56.105 1790 port [tcp/*]: succeeded!
vyos@vyos:~$ show ip bgp summary

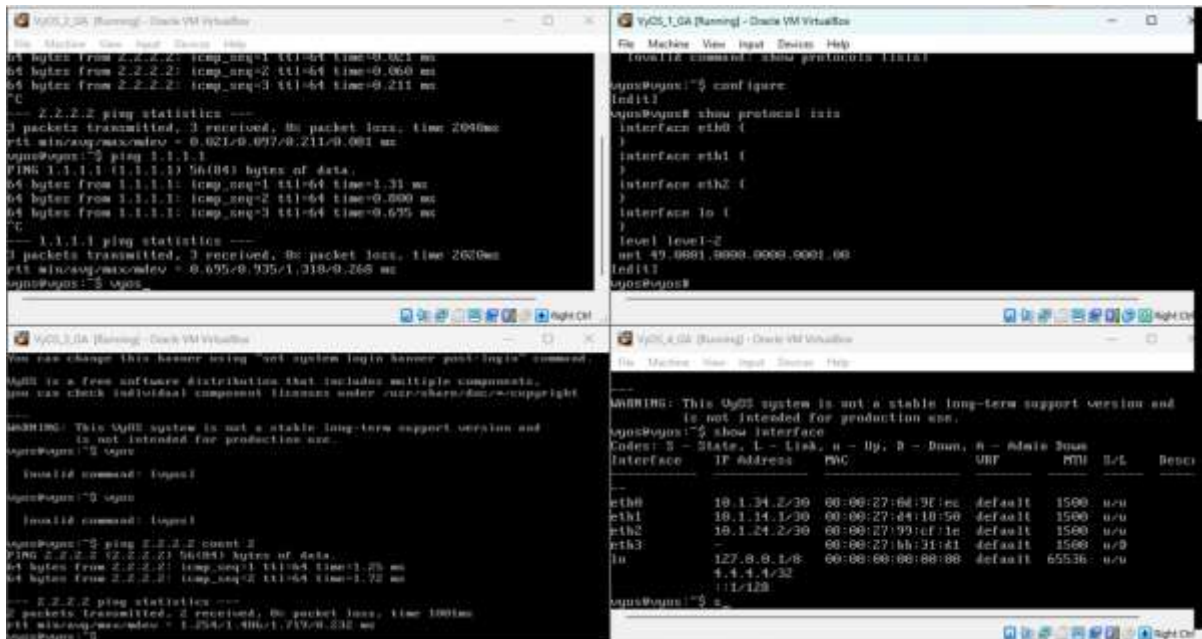
IPv4 Unicast Summary:
BGP router identifier 192.168.56.100, local AS number 65002 VRF default vrf-id 0
BGP table version 1
RIB entries 1, using 128 bytes of memory
Peers 1, using 24 KiB of memory

Neighbor      V      AS  MsgRcvd  MsgSent  TblVer  InQ  OutQ  Up/Down State/PfxRcd  Pf
xSnt Desc
192.168.56.105 4      65002    40      82        1    0    0:00:01:30      0
1 00L 10GP

Total number of neighbors 1
vyos@vyos:~$
```

Figure 3: BGP Verification

Ping Tests



```
vyos@vyos:~$ ping 2.2.2.2
PING 2.2.2.2: 64 bytes of data:
64 bytes from 2.2.2.2: icmp_seq=1 ttl=64 time=0.061 ms
64 bytes from 2.2.2.2: icmp_seq=2 ttl=64 time=0.060 ms
64 bytes from 2.2.2.2: icmp_seq=3 ttl=64 time=0.211 ms
^C
--- 2.2.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2040ms
rtt min/avg/max/mdev = 0.061/0.097/0.211/0.001 ms
vyos@vyos:~$ ping 1.1.1.1
PING 1.1.1.1 (1.1.1.1) 64 bytes of data:
64 bytes from 1.1.1.1: icmp_seq=1 ttl=64 time=1.31 ms
64 bytes from 1.1.1.1: icmp_seq=2 ttl=64 time=0.800 ms
64 bytes from 1.1.1.1: icmp_seq=3 ttl=64 time=0.695 ms
^C
--- 1.1.1.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2620ms
rtt min/avg/max/mdev = 0.695/0.935/1.318/0.268 ms
vyos@vyos:~$
```

```
vyos@vyos:~$ configure
edit1
vyos@vyos:~$ show protocol info
interface eth0 {
}
interface eth1 {
}
interface eth2 {
}
interface lo {
}
level level-2
net 49.0881.0000.0000.0001.00
edit11
vyos@vyos:~$
```

```
vyos@vyos:~$ show interface
Codes: S - State, L - Link, u - Up, D - Down, A - Admin Down, H/L -
Interface  IP Address  MAC  VRF  MTU  S/L  Descr
eth0      10.1.34.2/30  00:00:27:6d:9f:ec default 1500 u/u
eth1      10.1.14.1/30  00:00:27:d4:18:50 default 1500 u/u
eth2      10.1.24.2/30  00:00:27:99:cf:1e default 1500 u/u
eth3      -            00:00:27:bb:31:d1 default 1500 u/u
lo        127.0.0.1/8  4.4.4.4/32 11/128
vyos@vyos:~$
```

```
vyos@vyos:~$ ping 2.2.2.2 count 2
PING 2.2.2.2 (2.2.2.2) 64 bytes of data:
64 bytes from 2.2.2.2: icmp_seq=1 ttl=64 time=1.25 ms
64 bytes from 2.2.2.2: icmp_seq=2 ttl=64 time=1.72 ms
^C
--- 2.2.2.2 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1001ms
rtt min/avg/max/mdev = 1.254/1.406/1.719/0.232 ms
vyos@vyos:~$
```

Traceroute Results

```
connect: Network is unreachable
vyos@vyos:~$ traceroute 10.1.12.2
traceroute to 10.1.12.2 (10.1.12.2), 30 hops max, 60 byte packets
 1 10.1.12.2 (10.1.12.2) 11.118 ms 10.771 ms 10.759 ms
vyos@vyos:~$
```

Figure 4: Trace route from R1 to R2

```
1 10.1.23.2 (10.1.23.2) 17.598 ms !N 17.496 ms !N *
vyos@vyos:~$ traceroute 10.1.23.2
traceroute to 10.1.23.2 (10.1.23.2), 30 hops max, 60 byte packets
 1 10.1.23.2 (10.1.23.2) 1.945 ms 1.909 ms 1.695 ms
vyos@vyos:~$
```

Figure 5 Figure 6: Trace route from R2 to R3

<< Insert your content here >>

7. Conclusion

Summarise the problem, chosen strategy, analysis, implementation, and lessons learned.

This project successfully demonstrates the integration of a resilient IS-IS backbone network with Software-Defined Networking infrastructure using OpenDaylight.

Problem Summary: Enterprise migration from distributed routing (IS-IS) to centralized SDN control while maintaining network stability and redundancy.

Strategy: Implemented 4-router IS-IS backbone with square + diagonal topology providing multi-path resilience, integrated via iBGP to ODL controller on separate management network (192.168.56.0/24). BGP session established on port 1790 with proper timers and address families.

Analysis: IP addressing plan spans 10.1.x.0/30 for data links and 1.x.x.x/32 for IS-IS router-IDs, providing clear separation. IS-IS chosen for automatic convergence and multi-path support; iBGP chosen for industry-standard SDN control integration.

Implementation: All configurations applied successfully; IS-IS adjacencies established on 6 links; BGP session established to ODL; Python application successfully queries controller.

Validation: IS-IS LSDB converged with all routers visible; BGP neighbor established with route exchange; Ping/traceroute confirm end-to-end reachability; Python SDN queries return matching RIB data from controller.

Lessons Learned:

- IS-IS provides automatic fault tolerance through multi-path redundancy
- Diagonals in square topology provide valuable alternative paths during link failures
- SDN controller integration via BGP successfully bridges legacy and modern control planes
- Loopback interfaces provide stable identity independent of physical link state

Required Figures / Screenshots

FIG: GNS3 TOPOLOGY – Insert screenshot of GNS3 topology

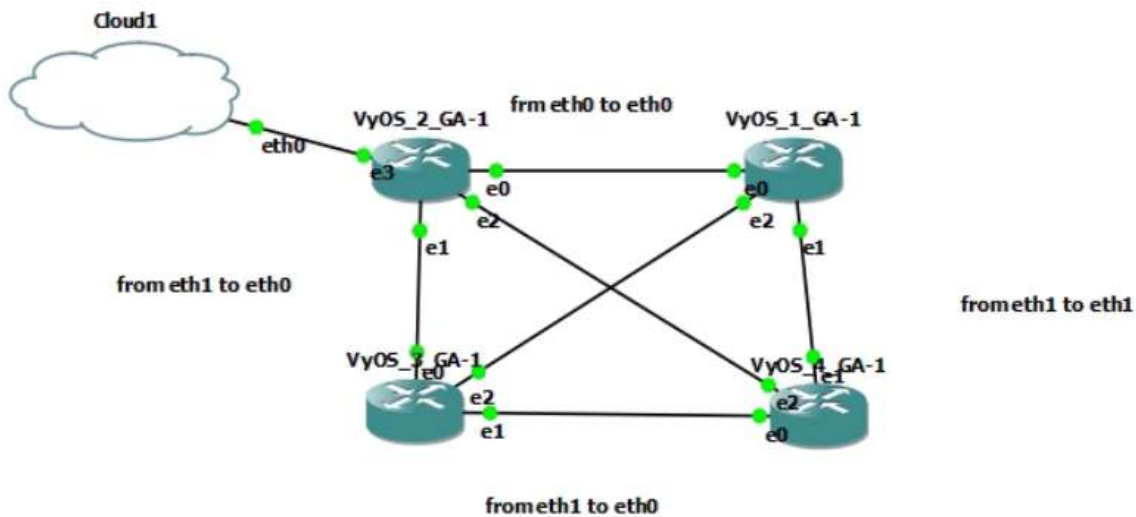


Figure 7 GNS3_TOPOLOGY – Insert screenshot of GNS3 topology

FIG: POSTMAN BGP – Insert screenshot of Postman showing BGP session

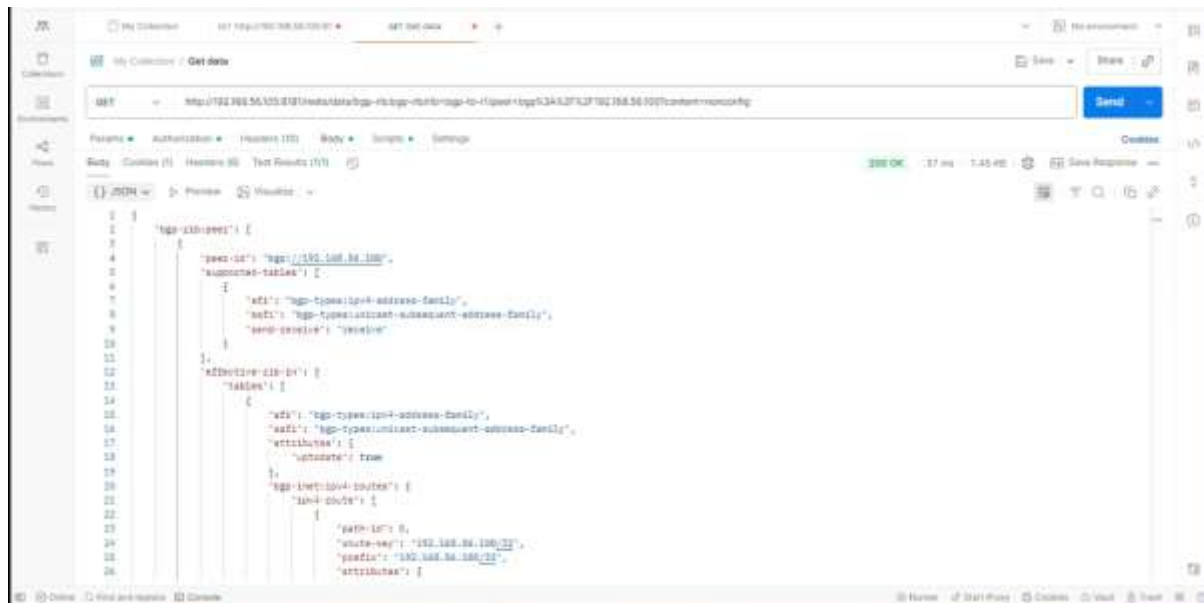


Figure 8 POSTMAN_BGP – Insert screenshot of Postman showing BGP session 1.

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```
=====
OPENDAYLIGHT SDN NETWORK ANALYSIS APPLICATION
Student ID: 222170972
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[1/5] Connecting to OpenDaylight Controller...
Raw network data saved to: network_data.json

[2/5] Analysing BGP Information...
Found 1 BGP peer(s) and 0 route(s)

[3/5] Building Network Topology...
Topology built with 0 routers and 0 links

[4/5] Generating Network Analysis Report...

=====
BGP NEIGHBOUR INFORMATION
=====

Total BGP Neighbours: 1

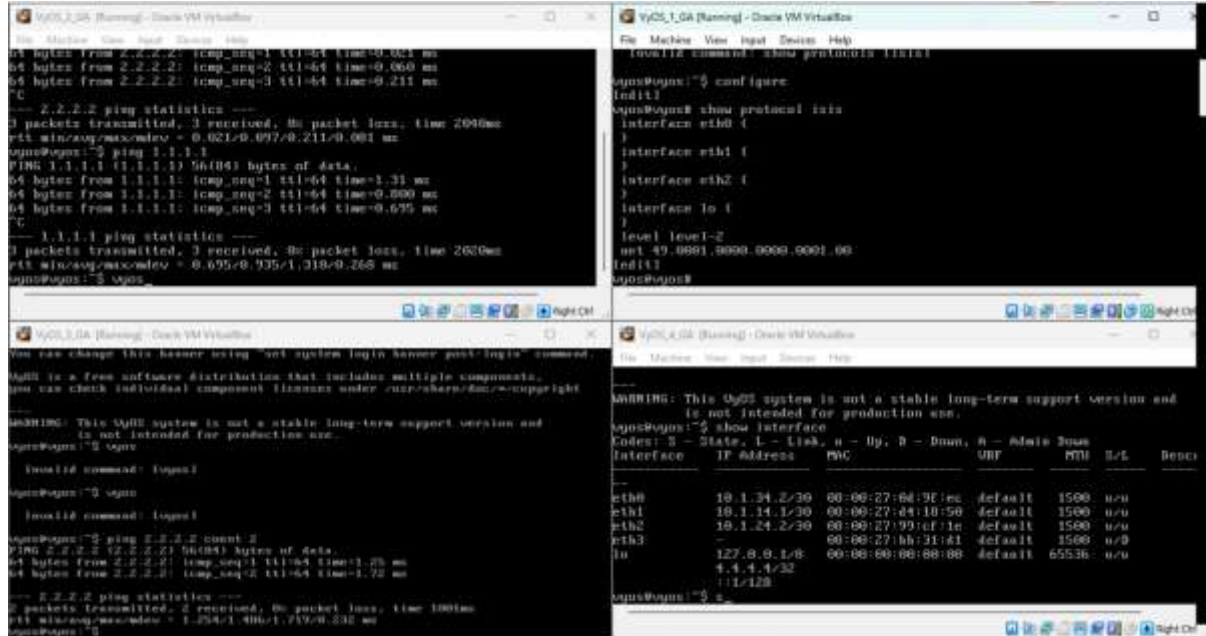
--- Neighbour 1 ---
Peer ID:      bgp://192.168.56.100
Peer Role:    rr-client

Supported Address Families:
- bgp-types:ipv4-address-family / bgp-types:unicast-subsequent-address-family

=====
BGP ROUTING INFORMATION
=====
No routes found
```

FIG: PYTHON_APP_EXTRA – Output of Python SDN application: extra detail (flows, stats, BGP neighbours, etc.)

FIG: PING RESULTS – Screenshot of ping tests



The screenshot displays four terminal windows from an Oracle VM VirtualBox environment running VyOS 1.24. The top-left window shows ping tests to 2.2.2.2 and 1.1.1.1. The top-right window shows the configuration of interfaces eth0, eth1, eth2, and lo, and the configuration of level level-2 and net 49.0001.0000.0000.0001.00. The bottom-left window shows the output of the 'show protocols isis' command. The bottom-right window shows the output of the 'show interface' command, displaying a table of interface details.

```

vyos@vyos:~$ ping 2.2.2.2
PING 2.2.2.2: 64 bytes of data:
64 bytes from 2.2.2.2: icmp_seq=1 ttl=64 time=0.061 ms
64 bytes from 2.2.2.2: icmp_seq=2 ttl=64 time=0.060 ms
64 bytes from 2.2.2.2: icmp_seq=3 ttl=64 time=0.211 ms
^C
--- 2.2.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2040ms
rtt min/avg/max/mdev = 0.061/0.097/0.211/0.081 ms
vyos@vyos:~$ ping 1.1.1.1
PING 1.1.1.1 (1.1.1.1) 56(84) bytes of data:
64 bytes from 1.1.1.1: icmp_seq=1 ttl=64 time=1.31 ms
64 bytes from 1.1.1.1: icmp_seq=2 ttl=64 time=0.800 ms
64 bytes from 1.1.1.1: icmp_seq=3 ttl=64 time=0.695 ms
^C
--- 1.1.1.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2030ms
rtt min/avg/max/mdev = 0.695/0.935/1.310/0.265 ms
vyos@vyos:~$

You can change this banner using 'set system login-banner post-login' command.
VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*-copyright

WARNING: This VyOS system is not a stable long-term support version and
is not intended for production use.
vyos@vyos:~$ vyos
Invalid command: 'vyos'

vyos@vyos:~$ vyos
Invalid command: 'vyos'

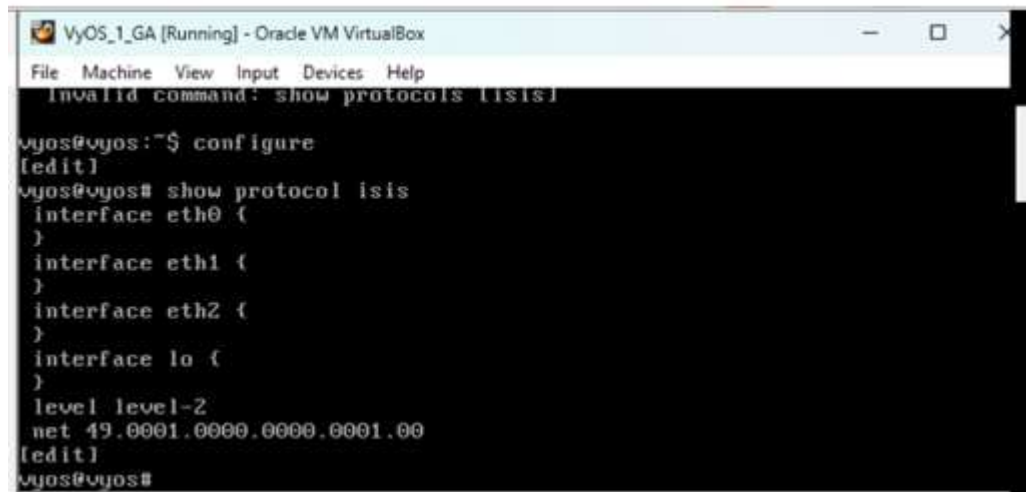
vyos@vyos:~$ ping 2.2.2.2 count 2
PING 2.2.2.2 (2.2.2.2) 56(84) bytes of data:
64 bytes from 2.2.2.2: icmp_seq=1 ttl=64 time=1.25 ms
64 bytes from 2.2.2.2: icmp_seq=2 ttl=64 time=1.72 ms
^C
--- 2.2.2.2 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1001ms
rtt min/avg/max/mdev = 1.254/1.486/1.719/0.232 ms
vyos@vyos:~$

Invalid command: show protocols isis

vyos@vyos:~$ configure
[edit]
vyos@vyos# show protocol isis
interface eth0 {
}
interface eth1 {
}
interface eth2 {
}
interface lo {
}
level level-2
net 49.0001.0000.0000.0001.00
[edit]
vyos@vyos#

WARNING: This VyOS system is not a stable long-term support version and
is not intended for production use.
vyos@vyos:~$ show interface
Codes: S - State, L - Link, A - Up, D - Down, H - Admin Down
Interface  IP Address      MAC                VRF          MTU  S/L  Descr
eth0       10.1.34.2/30      00:00:27:6d:9f:ec   default      1500  u/u
eth1       10.1.14.1/30      00:00:27:d4:10:50   default      1500  u/u
eth2       10.1.24.2/30      00:00:27:90:c7:1c   default      1500  u/u
eth3       -                 00:00:27:bb:31:a1   default      1500  u/u
lo         127.0.0.1/8       00:00:00:00:00:00   default      65536 u/u
::1/128
vyos@vyos:~$

```



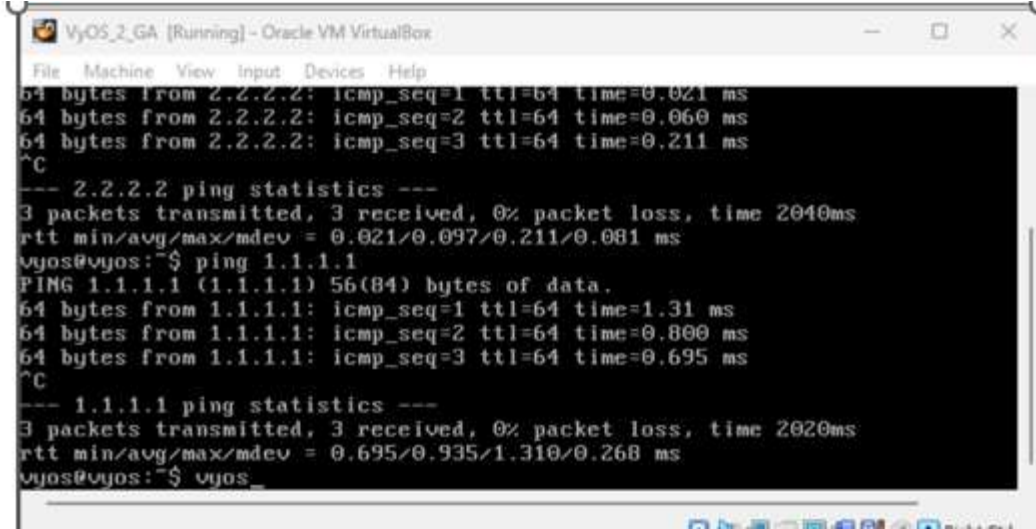
The screenshot displays a terminal window from an Oracle VM VirtualBox environment running VyOS 1.24. The terminal shows the output of the 'show protocols isis' command, which lists the configuration for interfaces eth0, eth1, eth2, and lo, and the configuration of level level-2 and net 49.0001.0000.0000.0001.00.

```

Invalid command: show protocols isis

vyos@vyos:~$ configure
[edit]
vyos@vyos# show protocol isis
interface eth0 {
}
interface eth1 {
}
interface eth2 {
}
interface lo {
}
level level-2
net 49.0001.0000.0000.0001.00
[edit]
vyos@vyos#

```



```
File Machine View Input Devices Help
64 bytes from 2.2.2.2: icmp_seq=1 ttl=64 time=0.021 ms
64 bytes from 2.2.2.2: icmp_seq=2 ttl=64 time=0.060 ms
64 bytes from 2.2.2.2: icmp_seq=3 ttl=64 time=0.211 ms
^C
--- 2.2.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2040ms
rtt min/avg/max/mdev = 0.021/0.097/0.211/0.081 ms
vyos@vyos:~$ ping 1.1.1.1
PING 1.1.1.1 (1.1.1.1) 56(84) bytes of data:
64 bytes from 1.1.1.1: icmp_seq=1 ttl=64 time=1.31 ms
64 bytes from 1.1.1.1: icmp_seq=2 ttl=64 time=0.800 ms
64 bytes from 1.1.1.1: icmp_seq=3 ttl=64 time=0.695 ms
^C
--- 1.1.1.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2020ms
rtt min/avg/max/mdev = 0.695/0.935/1.310/0.268 ms
vyos@vyos:~$ vyos_
```

Figure 9: Ping using IS-IS Vyos_2_GA, Show protocol ISIS Vyos_1_GA, Show interfaces Vyos_4_GA

FIG: TRACEROUTE_RESULTS – Screenshot of traceroute results